

DECK 03 · WEEK 2 · MORNING 1

Sampling, Hypothesis Testing & Regression

From a sample to a population — the inferential toolkit underlying every empirical claim in finance

CFA Level I · Vol 1 Learning Modules 7–11

Roadmap — Sampling, Hypothesis Testing & Simple Linear Regression

6 sections · ~2 hours · Built bottom-up from CFA readings

01

Sampling distributions & CLT

Probability vs non-probability sampling · CLT · standard error · bootstrap & jackknife

02

Confidence intervals

Reliability factors · t vs z · Sendar Fund 95% CI · what the interval does NOT mean

03

Hypothesis-testing framework

Six-step process · Type I / Type II · power · p-values · two-tailed t-test case study

04

Common tests (t, x^2 , F, ρ)

Cheat sheet · single variance · variance ratio · correlation (Pearson & Spearman) · independence

05

Simple linear regression

OLS · four assumptions · ANOVA & R^2 · coefficient tests · prediction intervals · functional forms

06

Big data & ML wrap

Three Vs + veracity · supervised / unsupervised / deep learning · overfitting · train-validate-test

Learning objectives

By the end of this deck you should be able to:

1. Choose between simple random, systematic, stratified, cluster, convenience and judgmental sampling, and compute the standard error of a sample mean using the central limit theorem.
2. Build a confidence interval for a population mean using the correct reliability factor, and state precisely what the interval does and does not claim.
3. Run the six-step hypothesis-testing process, distinguishing Type I from Type II errors and one-tailed from two-tailed critical values.
4. Apply the right test statistic — t , z , chi-square or F — to a question about means, variances, correlations or categorical independence, and know its degrees of freedom.
5. Estimate a simple linear regression, test whether its slope differs from zero (or from any hypothesised value), and interpret R^2 , the standard error of estimate, prediction intervals and log functional forms — then place all of it inside the big-data / ML landscape.

SECTION 01

Sampling distributions & CLT

Probability sampling gives every element a known chance of selection; non-probability sampling buys speed and cost at the risk of a non-representative sample

All else equal, probability methods yield more accuracy and reliability; non-probability methods trade precision for speed and cost.

Method	Type	How it works	Best when
Simple random	Probability	Every element equally likely	Homogeneous population
Systematic	Probability	Take every kth element in a list	Population cannot be listed
Stratified random	Probability	Sample each stratum pro rata	Indexing, known risk factors
Cluster (one/two stage)	Probability	Pick clusters, then sample within	Broad geographic surveys
Convenience	Non-probability	Take the cheapest data to access	Pilot studies, cost limits
Judgmental	Non-probability	An expert hand-picks the sample	Auditing, specialist tasks

WHY STRATIFIED BEATS SIMPLE RANDOM IN FIXED INCOME

Full replication of a bond index means owning thousands of illiquid issues. Stratify instead: split the universe by duration, cash-flow profile, sector, credit quality and call exposure, then sample each cell in proportion to its index weight. Same risk exposures, a fraction of the line items — and smaller estimation variance than a simple random draw.

¹ For a given sample size, cluster sampling is usually the LEAST accurate probability method — but the most time- and cost-efficient.

Source: CFA Institute, Curriculum Vol 1, Learning Module 7, §2 Sampling Methods, pp. 197–201.

Whatever shape the population has, the sample mean goes approximately Normal(μ , σ^2/n) once n is large — $n \geq 30$ is the working rule of thumb

The CLT is the bedrock of inference: it lets us point normal-distribution tooling at a sample mean even when the underlying returns are nowhere near normal.

$$\bar{X} \approx N(\mu, \sigma^2 / n) \quad \text{for large } n$$

where: μ = population mean; σ^2 = population variance (must be FINITE); n = sample size. Requires a random sample; says nothing about any single observation.

SHAPE

Skewed, bimodal or heavy-tailed population — the distribution of the sample MEAN still converges to a bell curve. If the population is badly non-normal, n well above 30 may be needed.

CENTRE

The sampling distribution is centred on μ : the sample mean is an UNBIASED estimator. More data does not move the centre — it tightens the spread.

SPREAD

Variance shrinks by $1/n$, so standard deviation shrinks by $1/\sqrt{n}$. That standard deviation has its own name: the STANDARD ERROR of the mean.

Standard error falls like $1/\sqrt{n}$ – quadrupling the sample only halves the error, so precision gets expensive fast

CFA Example 2: 1,258 daily returns of the EAA Equity Index are the "population" (mean 0.035%). Samples are drawn 100 times at each size and the mean absolute error recorded.

$$\sigma_{\bar{X}} = \sigma/\sqrt{n} \quad (\sigma \text{ known})$$

$$s_{\bar{X}} = s/\sqrt{n} \quad (\sigma \text{ unknown})$$

where: $s^2 = \sum(X_i - \bar{X})^2 / (n-1)$. The $n-1$ denominator is what makes s^2 an unbiased estimator of σ^2 . In practice we almost always use s .

Sample size n	Random MAE	Stratified MAE	Ratio vs n = 5	Theory $1/\sqrt{5/n}$
5	0.297%	0.294%	1.00×	1.00×
20	0.163%	0.152%	0.55×	0.50×
50	0.091%	0.083%	0.31×	0.32×
100	0.063%	0.071%	0.21×	0.22×
200	0.039%	0.051%	0.13×	0.16×
500	0.019%	0.025%	0.06×	0.10×
1,000	0.009%	0.008%	0.03×	0.07×

Source: CFA Institute, Curriculum Vol 1, Learning Module 7, Example 2 (EAA Equity Index), Exhibit 2, pp. 201–202.

Bootstrap and jackknife estimate a sampling distribution from the data itself — the tool of choice when no analytical standard-error formula exists

Both are model-free: no assumption about the shape of the underlying population, and no closed-form statistic required.

BOOTSTRAP · DRAW WITH REPLACEMENT

Draw B resamples, each of the SAME size n as the original sample, sampling with replacement. Compute the statistic on each. The spread of those B values approximates the true sampling distribution.

B is your choice. Results differ run to run.

Use for: standard error or CI of a median, a Sharpe ratio, a drawdown — anything with no formula.

JACKKNIFE · LEAVE ONE OUT

Remove one observation at a time and recompute the statistic on the remaining $n - 1$. Exactly n repetitions — no more, no fewer.

Deterministic: same answer every run.

Use for: reducing the BIAS of an estimator, plus standard errors and confidence intervals.

CFA EXAMPLE 4 · BOOTSTRAPPING A THINLY TRADED STOCK

12 monthly returns, B = 1,000 resamples. Mean across all resample means = -0.01367 ; $\sum(\hat{\theta}_b - \bar{\theta})^2 = 1.94143$. Standard error = $\sqrt{1.94143 / 999} = 0.04408$, i.e. 4.4% — computed without ever writing down $\sigma/\sqrt{n}$.

Source: CFA Institute, Curriculum Vol 1, Learning Module 7, §4 Resampling, Example 4, pp. 209–212.

SECTION 02

Confidence intervals

A confidence interval is the point estimate plus and minus a reliability factor times the standard error — t when σ is unknown, z when σ is known or n is very large

In finance σ is essentially never known, so the t -distribution is the default. As df grows, t collapses onto z — look down the last column.

$$CI = \bar{X} \pm t_{\{\alpha/2, n-1\}} \cdot s/\sqrt{n} \quad (z_{\{\alpha/2\}} \text{ if } \sigma \text{ known })$$

where: $\bar{X}$ = sample mean; s = sample standard deviation; n = sample size. The reliability factor depends on the confidence level AND on the degrees of freedom.

Confidence level	α	$z_{\{\alpha/2\}}$	t (df = 10)	t (df = 30)	t (df = 120)
90%	0.10	1.645	1.812	1.697	1.658
95%	0.05	1.960	2.228	2.042	1.980
99%	0.01	2.576	3.169	2.750	2.617

WHAT A 95% INTERVAL DOES NOT MEAN

It does NOT mean "there is a 95% probability that μ lies between these two numbers". μ is a fixed constant — it is either in there or it is not. It means: if we repeated the sampling exercise many times and built an interval each time, 95% of those intervals would contain μ . The randomness is in the interval, not in μ .

Source: CFA Institute, Curriculum Vol 1 — reliability factors from Learning Module 12, Appendix B (Student's t) and Appendix A (standard normal).

Sendar Fund's 95% interval for mean monthly return runs from -0.02% to $+3.02\%$ — wide enough to swallow the 1.10% model benchmark whole

Building a confidence interval and running a two-tailed test at the same α are two views of one calculation. Hold that thought — we prove it in §3.

SENDER EQUITY FUND — 95% CONFIDENCE INTERVAL FOR THE MEAN MONTHLY RETURN

Sendar Equity Fund, a mid-cap growth fund with a 24-month track record: $\bar{X} = 1.50\%$, $s = 3.60\%$. A pricing model says the fund should have earned 1.10% a month given its market risk. Build the 95% interval and see whether 1.10% sits inside it.

1. Degrees of freedom: $n - 1 = 24 - 1 = 23$.
2. Reliability factor, 95% two-sided, $df = 23$: $t^*_{0.025, 23} = 2.069$.
3. Standard error: $s / \sqrt{n} = 3.60 / \sqrt{24} = 3.60 / 4.8990 = 0.7348$.
4. Margin of error: $t^* \times SE = 2.069 \times 0.7348 = 1.5204$.
5. 95% CI: $1.50 \pm 1.5204 \rightarrow [-0.0204\%, +3.0204\%]$.
6. 1.10% lies INSIDE the interval, so the data cannot distinguish the fund's mean from the model expectation at 5% significance.

95% CI = [-0.02% , $+3.02\%$] · 1.10% is inside · no evidence of outperformance.

Source: CFA Institute, Curriculum Vol 1, Learning Module 8, §3 Tests of Return and Risk, Exhibit 5, pp. 224–225.

SECTION 03

Hypothesis-testing framework

Every hypothesis test is the same six steps — state, choose, set α , write the rule, compute, decide

H_0 and H_a are statements about a POPULATION PARAMETER, and they must be mutually exclusive and collectively exhaustive. The "hoped-for" or suspected condition goes in H_a .

1 State the hypotheses

Write H_0 and H_a in terms of the population parameter. Two-sided ($=$ vs $\neq$) or one-sided ($\leq$ vs $>$, $\geq$ vs $<$).

2 Identify the test statistic

Driven entirely by what you are testing: t for a mean, χ^2 for a variance, F for a variance ratio, χ^2 for independence.

3 Specify the significance level

α = the probability of a Type I error you are willing to run. Convention: 10%, 5%, 1%. Lower α = harder to reject.

4 State the decision rule

Look up the critical value(s) for this α , this df AND this number of tails. Reject if the statistic is MORE EXTREME.

5 Calculate the test statistic

Plug the sample into the formula. Report the value and, wherever software allows, the p-value alongside it.

6 Make a decision

Reject H_0 if the statistic is beyond the critical value — equivalently, if $p \leq \alpha$. Otherwise, fail to reject.

¹ We never "accept" H_0 — only reject or fail to reject. Failing to reject means weak evidence, not a true null.

Source: CFA Institute, Curriculum Vol 1, Learning Module 8, §2 The Process of Hypothesis Testing, Exhibit 1, pp. 219–222.

Type I rejects a true null (false positive, probability α); Type II fails to reject a false null (false negative, probability β) — and lowering one raises the other

The ONLY way to reduce both error probabilities at the same time is to increase the sample size n . Everything else is a trade.

Your decision	H_0 is TRUE	H_0 is FALSE
Fail to reject H_0	Correct decision $(1 - \alpha)$	TYPE II — false negative (β)
Reject H_0	TYPE I — false positive (α)	Correct — POWER $(1 - \beta)$

WHICH ERROR HURTS MORE?

It depends entirely on the consequences, and that is what should set α . Approving a drug that does not work is a costly Type I. Missing a borrower who then defaults is a costly Type II. Choose α from the cost of being wrong, not from habit.

POWER = 1 - β

Power = $P(\text{reject } H_0 \mid H_0 \text{ is false})$ — the ability to detect a real effect. It rises with a larger true effect, a larger n , a smaller σ , and a higher α . An underpowered study is worse than no study: it produces a confident "no effect".

The p-value is the smallest α at which the null could be rejected — reject when $p \leq \alpha$, and never read it as the probability that H_0 is true

A two-tailed p-value is TWICE the one-tail probability beyond $|t_{\text{calc}}|$. Reporting p is strictly more informative than reporting a binary reject / fail-to-reject.

$$p = P(|T| \geq |t_{\text{calc}}| \mid H_0 \text{ true})$$

Reject H_0 iff $p \leq \alpha$

where: T = the test-statistic random variable under H_0 ; t_{calc} = the value realised from this sample. p is a statement about the DATA given H_0 — not about H_0 given the data.

TWO - TAILED

$H_0: \theta = \theta_0$ vs $H_a: \theta \neq \theta_0$
Reject if $|t_{\text{calc}}| > t_{\{\alpha/2, \text{df}\}}$
 $p = 2 \times P(T > |t_{\text{calc}}|)$
 α is SPLIT across both tails.

RIGHT - TAILED

$H_0: \theta \leq \theta_0$ vs $H_a: \theta > \theta_0$
Reject if $t_{\text{calc}} > t_{\{\alpha, \text{df}\}}$
 $p = P(T > t_{\text{calc}})$
All of α sits in the upper tail.

LEFT - TAILED

$H_0: \theta \geq \theta_0$ vs $H_a: \theta < \theta_0$
Reject if $t_{\text{calc}} < -t_{\{\alpha, \text{df}\}}$
 $p = P(T < t_{\text{calc}})$
All of α sits in the lower tail.

Sender's t-statistic of 0.544 sits well inside the ± 2.069 band, so at 5% we fail to reject the 1.10% benchmark — exactly what the confidence interval already told us

Same fund, same numbers, same α as slide 11 — and necessarily the same conclusion. A two-tailed test at α and a $100(1-\alpha)\%$ interval are one calculation in two costumes.

THE SIX STEPS APPLIED — t-TEST OF A SINGLE MEAN

Sender Equity Fund: $\bar{X} = 1.50\%$, $s = 3.60\%$, $n = 24$ months. Test at $\alpha = 5\%$ whether the true mean monthly return differs from the 1.10% implied by the pricing model. Returns assumed approximately normal.

1. Hypotheses. $H_0: \mu = 1.10\%$ vs $H_a: \mu \neq 1.10\%$ — two-tailed, because "differs" has no direction.
2. Test statistic. $t = (\bar{X} - \mu_0) / (s/\sqrt{n})$, t-distributed with $df = n - 1 = 23$. (σ unknown $\rightarrow t$, not z .)
3. Significance. $\alpha = 5\%$, split as 2.5% in each tail.
4. Decision rule. Reject if $t_{\text{calc}} < -2.069$ or $t_{\text{calc}} > +2.069$.
5. Compute. $t = (1.50 - 1.10) / (3.60/\sqrt{24}) = 0.40 / 0.7348 = 0.5443$.
6. Decide. $|0.5443| < 2.069 \rightarrow$ fail to reject H_0 (two-tailed $p \approx 0.59$, nowhere near 0.05).

Fail to reject H_0 — insufficient evidence that the mean differs from 1.10%. NOT "the fund earns 1.10%".

SECTION 04

Common tests (t, z, χ^2 , F, ρ)

One table maps every question about means, variances, correlations or categories to its test statistic and degrees of freedom

Step 1 of every test is always the same question: which row of this table matches what I am asking? Everything after that is mechanical.

What we test	Test statistic	Distribution	Degrees of freedom
Single mean (σ unknown)	$t = (\bar{X} - \mu_0) / (s/\sqrt{n})$	t	$n - 1$
Difference in means (independent)	$t = (\bar{X}_1 - \bar{X}_2) / \sqrt{(s^2_p/n_1 + s^2_p/n_2)}$	t	$n_1 + n_2 - 2$
Mean of paired differences	$t = (\bar{d} - \mu_{d0}) / s_{\bar{d}}$	t	$n - 1$
Single variance	$\chi^2 = (n - 1)s^2 / \sigma_0^2$	χ^2	$n - 1$
Equality of two variances	$F = s^2_1 / s^2_2$	F	$n_1 - 1, n_2 - 1$
Correlation equals zero	$t = r \sqrt{(n-2)} / \sqrt{(1 - r^2)}$	t	$n - 2$
Independence of two categoricals	$\chi^2 = \sum (O_{ij} - E_{ij})^2 / E_{ij}$	χ^2	$(r-1)(c-1)$

¹ Pooled variance $s^2_p = [(n_1-1)s^2_1 + (n_2-1)s^2_2] / (n_1+n_2-2)$. Assumes independent samples, normal populations, equal variances.

² Paired data (same days, same companies)? Use the mean-of-differences test — more powerful, because it strips out shared variation.

Source: CFA Institute, Curriculum Vol 1, Learning Module 8, Exhibit 2, p. 221; Learning Module 9, §2–§3, pp. 246–256.

Chi-square tests a single variance against a hypothesised level — and because the distribution is asymmetric, the two tails have completely different critical values

Unlike t and z , chi-square is skewed and lives on $[0, \infty)$. You cannot flip the sign of a critical value to get the other tail — you must look up each one separately.

$$\chi^2 = (n - 1) \cdot s^2 / \sigma_0^2 \qquad \text{df} = n - 1$$

where: s^2 = sample variance; σ_0^2 = hypothesised population variance. Assumes the underlying population is normally distributed — this test is not robust to that.

SENDER FUND — IS TRUE VOLATILITY BELOW THE 4% TRIGGER?

Same fund: $n = 24$, $s = 3.60\%$ so $s^2 = 12.96$. A mandate trigger sits at $\sigma = 4\%$, i.e. $\sigma^2 = 16$. Test at $\alpha = 5\%$.

1. $H_0: \sigma^2 \geq 16$ vs $H_a: \sigma^2 < 16$ — one-sided, LEFT tail (the suspected condition goes in H_a).
2. $\chi^2_{\text{calc}} = (24 - 1) \times 12.96 / 16 = 23 \times 0.81 = 18.63$, with $\text{df} = 23$.
3. Critical value with 23 df and 5% in the LEFT tail = 13.091.
4. Decision rule: reject H_0 only if $\chi^2_{\text{calc}} < 13.091$.
5. Decide: $18.63 > 13.091 \rightarrow$ fail to reject.

Fail to reject — no evidence that true volatility is below 4%, even though observed $s = 3.60\%$.

The F-test compares two variances as a ratio — and the same F of 1.185 fails two-tailed but rejects one-tailed, which is why the tail count is never a detail

Modern practice does not force the larger variance on top: software returns a lower AND an upper critical value, and either tail can reject. Forming the ratio the other way up gives the same conclusion.

$$F = s^2_1 / s^2_2$$

$$df = (n_1 - 1, n_2 - 1)$$

where: Numerator df comes from sample 1, denominator df from sample 2. F is right-skewed on $[0, \infty)$. Both populations assumed normal — the F-test is sensitive to that assumption.

DID A REGULATORY CHANGE ALTER INDEX VOLATILITY? • CFA EXAMPLE 3

418 weekly returns before a market regulation change (variance 4.644) and 418 weeks after (variance 3.919). $\alpha = 5\%$, $df = (417, 417)$.
 $F_{\text{calc}} = 4.644 / 3.919 = 1.185$ either way you ask the question.

1. TWO-TAILED — $H_0: \sigma^2_{\text{before}} = \sigma^2_{\text{after}}$. Critical values 0.8251 and 1.2119. 1.185 falls inside → FAIL TO REJECT.
2. ONE-TAILED — $H_0: \sigma^2_{\text{before}} \leq \sigma^2_{\text{after}}$. Single critical value 1.1750. $1.185 > 1.1750$ → REJECT.
3. Identical data, identical statistic, opposite conclusions — the whole difference is where you put the 5%.
4. The tail count is fixed by the QUESTION ("different?" vs "greater?"), and must be fixed before you see the data.
5. Practitioner use: this is also the check on the equal-variance assumption behind a pooled t-test.

Two-tailed: no evidence of a change. One-tailed: volatility was significantly higher before. Choose the tails first.

Test a correlation with a t-statistic on $n-2$ degrees of freedom — and when the data are far from normal, run the same test on RANKS instead

The required $|r|$ to reach significance collapses as n grows: $r = 0.35$ is insignificant at $n = 12$ ($t = 1.18$) and significant at $n = 32$ ($t = 2.05$). At big-data sample sizes, almost everything "rejects".

$$t = r \sqrt{(n-2)} / \sqrt{(1 - r^2)}$$

$$r_s = 1 - 6 \sum d_i^2 / [n(n^2 - 1)]$$

where: Left: the t-test on a Pearson correlation, $df = n - 2$. Right: the Spearman rank correlation — run the SAME t-test on r_s once $n > 30$.

PEARSON • PARAMETRIC

Annual returns on Investment One and Investment Two over 33 years. Sample correlation $r = 0.43051$. Is it significantly POSITIVE at $\alpha = 1\%$?

1. $H_0: \rho \leq 0$ vs $H_a: \rho > 0$ — one-sided, right tail.
2. $df = 33 - 2 = 31$; critical value = 2.4528.
3. $t = 0.43051 \times \sqrt{31} / \sqrt{(1 - 0.18534)} = 2.6557$.
4. $2.6557 > 2.4528 \rightarrow$ reject H_0 .

Significantly positive correlation at 1%.

SPEARMAN • NON-PARAMETRIC

Funds A and B over 35 years, but the returns depart badly from normality. Rank each series, take rank differences: $\sum d^2 = 2,202$.

1. $r_s = 1 - 6(2,202) / [35 \times (1,225 - 1)] = 0.6916$.
2. $H_0: r_s = 0$ vs $H_a: r_s \neq 0$; $df = 33$, critical ± 2.0345 .
3. $t = 0.6916 \times \sqrt{33} / \sqrt{(1 - 0.6916^2)} = 5.5005$.
4. $|5.5005| > 2.0345 \rightarrow$ reject H_0 .

Rank correlation is significantly non-zero.

Source: CFA Institute, Curriculum Vol 1, Learning Module 9, §2 Tests Concerning Correlation, Examples 1–2 and Exhibit 4, pp. 246–252.

A chi-square test of independence compares observed counts with the counts you would expect if the two classifications were unrelated — always a right-tail test

Expected frequency under independence: $E_{ij} = (\text{row total} \times \text{column total}) / \text{grand total}$. Degrees of freedom = $(\text{rows} - 1) \times (\text{columns} - 1)$ — never $r \times c$.

Style	Small	Medium	Large	Total
Value	50	110	343	503
Growth	42	122	202	366
Blend	56	149	520	725
Total	148	381	1,065	1,594

Observed frequencies: 1,594 ETFs cross-tabulated by market-cap size (small / medium / large) and investment style (value / growth / blend). A 3×3 contingency table, so $m = 9$ cells and $df = 2 \times 2 = 4$.

Why chi-square and not correlation? Style is a category, not a number — there is no covariance to compute.

TEST OF INDEPENDENCE · $\alpha = 5\%$

H_0 : size and style are independent. H_a : they are related. One-sided by construction: deviations of either sign are squared, so the statistic can only push right.

1. Expected: $E_{ij} = (\text{row} \times \text{column}) / 1,594$. Small-value: $503 \times 148 / 1,594 = 46.70$.
2. Cell contribution: $(O - E)^2 / E$. Small-value: $(50 - 46.70)^2 / 46.70 = 0.233$.
3. Sum all 9 cells: $\chi^2_{\text{calc}} = 32.080$.
4. Critical value, $df = 4$, 5% right tail = 9.4877.
5. $32.080 > 9.4877 \rightarrow$ reject H_0 .

Size and investment style are related, not independent.

¹ Standardized (Pearson) residual = $(O - E) / \sqrt{E}$ locates the dependence: medium-cap growth is +3.69, large-cap growth is -2.72.

Source: CFA Institute, Curriculum Vol 1, Learning Module 9, §3 Contingency Table Data, Exhibits 7–9, pp. 255–258.

SECTION 05

Simple linear regression

Simple linear regression fits $Y = b_0 + b_1X + \varepsilon$ by minimising the sum of SQUARED residuals — the least-squares criterion

b_0 = intercept, the fitted value of Y when $X = 0$. b_1 = slope, the change in Y per one-unit change in X . Residual $e_i = Y_i - \hat{Y}_i$, in the same units as Y .

$$\hat{b}_1 = \Sigma(X_i - \bar{X})(Y_i - \bar{Y}) / \Sigma(X_i - \bar{X})^2$$

$$\hat{b}_0 = \bar{Y} - \hat{b}_1\bar{X}$$

$$\min \Sigma e_i^2$$

where: $\hat{b}_1$ is the covariance of X and Y divided by the variance of X . The denominators cancel, so only the two sums are needed. Both estimators are closed-form — no search, no iteration.

DEPENDENT vs INDEPENDENT

Y (dependent, explained) on the vertical axis; X (independent, explanatory) on the horizontal. Theory picks X , not the software. Swap them and you get a different line — regression is not symmetric, and it is not evidence of causation.

ERROR vs RESIDUAL

ERROR ε : unobservable, measured from the TRUE population line.

RESIDUAL e : observable, measured from the FITTED sample line.

Least squares forces $\Sigma e = 0$ by construction, so a zero mean residual proves nothing about fit.

Source: CFA Institute, Curriculum Vol 1, Learning Module 10, §2 Estimation of the SLR Model, pp. 267–273.

Regressing ROA on CAPEX across six companies gives $\hat{Y} = 4.875 + 1.25X$ — one extra point of CAPEX buys 1.25 points of ROA on average

This dataset runs through the rest of the section — estimation here, then ANOVA, then the slope test, then the prediction interval. Same six companies throughout.

RETURN ON ASSETS REGRESSED ON SCALED CAPEX · n = 6 COMPANIES

Six companies. ROA (Y) averages 12.5%; CAPEX scaled by prior-period PP&E (X) averages 6.1%. From the cross-products table: $\Sigma(X_i - \bar{X})^2 = 122.64$ and $\Sigma(Y_i - \bar{Y})(X_i - \bar{X}) = 153.30$, with $\Sigma(Y_i - \bar{Y})^2 = 239.50$.

1. Slope: $\hat{b}_1 = 153.30 / 122.64 = 1.25$.
2. Intercept: $\hat{b}_0 = \bar{Y} - \hat{b}_1\bar{X} = 12.5 - (1.25 \times 6.1) = 12.5 - 7.625 = 4.875$.
3. Fitted model: $\hat{Y}_i = 4.875 + 1.25 X_i$.
4. Read the slope: CAPEX up from 4% to 5% → predicted ROA up 1.25 percentage points.
5. Read the intercept: predicted ROA of 4.875% at zero CAPEX — meaningful only if $X = 0$ is a plausible state.
6. Correlation: $r = 30.66 / (6.9210 \times 4.9526) = 0.8945$. Note r and $\hat{b}_1$ always share a sign.

$\hat{Y} = 4.875 + 1.25 X$ · $r = 0.8945$ · significance still to be established.

¹ Same machinery, new labels: regress a stock on the market and $\hat{b}_1$ is beta, $\hat{b}_0$ is alpha. We test that regression on slide 28.

Source: CFA Institute, Curriculum Vol 1, Learning Module 10, §2 Estimation, Exhibits 2–7, pp. 268–275.

OLS rests on four assumptions — linearity, homoskedasticity, independence, normality — and the residual plot is where you go to check all four

Coefficients can survive mild violations. STANDARD ERRORS cannot — and standard errors are what every t-statistic, confidence interval and prediction interval is built from.

- Linearity**
1 $E[Y|X]$ is a straight line in X , and X is non-stochastic. Curvature in the residual plot means the model is biased — transform a variable (see slide 30) rather than shrug.
- Homoskedasticity**
2 Residual variance is constant across observations. A fan shape, or two visibly different regimes, signals heteroskedasticity — the CFA interest-rate example splits cleanly into normal-rate and low-rate eras.
- Independence**
3 Observations, and therefore residuals, are uncorrelated. In time series look for autocorrelation: the quarterly-revenue example spikes every Q4 and falls back — a pattern the model never sees.
- Normality**
4 The RESIDUALS are normally distributed — not X , not Y . Critical in small samples; in large samples the CLT lets you relax it. Outliers do more damage here than anything else.

Source: CFA Institute, Curriculum Vol 1, Learning Module 10, §3 Assumptions of the SLR Model, pp. 277–284.

Total variation splits into explained (SSR) plus unexplained (SSE) – R^2 is the explained share, and the F-test asks whether that share could have arisen by chance

In SIMPLE linear regression R^2 is just the squared correlation: $r = 0.8945 \rightarrow r^2 = 0.8001 = R^2$. And the F-test of the whole model is exactly the t-test on the single slope, squared.

$$SST = SSR + SSE$$

$$R^2 = SSR/SST$$

$$F = MSR/MSE$$

$$s_e = \sqrt{MSE}$$

where: $SST = \sum(Y - \bar{Y})^2$; $SSR = \sum(\hat{Y} - \bar{Y})^2$; $SSE = \sum(Y - \hat{Y})^2$. $MSR = SSR/1$; $MSE = SSE/(n-2)$. F has $df(1, n-2)$ and is always a RIGHT-tail test.

Source	Sum of squares	df	Mean square	F
Regression (SSR)	191.625	1	191.625 (MSR)	16.010
Error (SSE)	47.875	4	11.969 (MSE)	
Total (SST)	239.500	5		
R² = SSR/SST = 0.8001		s_e = √MSE = 3.4596		Critical F _{0.05, 1, 4} = 7.71
				Reject H ₀

Source: CFA Institute, Curriculum Vol 1, Learning Module 10, §4–§5 ANOVA and Measures of Fit, Exhibits 20 and 33, pp. 285–298.

The slope t-statistic tests b_1 against ANY hypothesised value on $n-2$ degrees of freedom — including the value that actually matters, $b_1 = 1$

Note the two tail counts below: five percent split two ways for a "different from" question, all five percent on one side for a "greater than" question.

$$t = (\hat{b}_1 - B_1) / s_{\hat{b}_1} \quad s_{\hat{b}_1} = s_e / \sqrt{\Sigma(X_i - \bar{X})^2} \quad df = n - 2$$

where: B_1 is whatever you hypothesise — zero is the default, not the only option. The more the independent variable varies, the smaller $s_{\hat{b}_1}$ and the sharper the test.

ROA ON CAPEX · IS THE SLOPE ZERO?

Continuing the six companies: $\hat{b}_1 = 1.25$, $s_e = 3.4596$, $\Sigma(X - \bar{X})^2 = 122.64$, $n = 6$. Test $H_0: b_1 = 0$ at $\alpha = 5\%$, two-tailed.

1. $s_{\hat{b}_1} = 3.4596 / \sqrt{122.64} = 3.4596 / 11.074 = 0.3124$.
2. $df = 6 - 2 = 4$; critical values ± 2.776 .
3. $t = (1.25 - 0) / 0.3124 = 4.0013$; $p = 0.016$.
4. Reject H_0 . Note $t^2 = 4.0013^2 = 16.010 =$ the F-statistic.

Slope significantly different from zero.

MARKET MODEL · IS BETA REALLY 1?

ABC daily returns regressed on the EAA equity index: $\hat{b}_1 = 0.982$, $s_e = 1.26$, $sd(X) = 0.70$, $n = 1,200$. Critical values ± 1.96 .

1. $\Sigma(X - \bar{X})^2 = 0.70^2 \times 1,199 = 587.51$; $s_{\hat{b}_1} = 1.26 / \sqrt{587.51} = 0.0520$.
2. Versus zero: $t = 0.982 / 0.0520 = 18.89 \rightarrow$ reject, emphatically.
3. Versus one: $t = (0.982 - 1) / 0.0520 = -0.35 \rightarrow$ fail to reject.
4. Beta is overwhelmingly non-zero and statistically indistinguishable from 1.0.

Significant ≠ different from one.

Source: CFA Institute, Curriculum Vol 1, Learning Module 10, §5 Hypothesis Tests of Coefficients, Exhibits 21 and 27, pp. 285–295.

A prediction interval widens the further X_f sits from the sample mean — forecast uncertainty is smallest at the sample's centre of gravity and grows in both directions

The interval carries BOTH the uncertainty in the fitted line and the irreducible residual noise — which is why the "1 +" sits inside the square root and why it is always wider than a confidence interval for the mean at X_f .

$$s_f = s_e \sqrt{1 + 1/n + (X_f - \bar{X})^2 / \sum (X_i - \bar{X})^2} \quad \text{PI} = \hat{Y}_f \pm t^* \cdot s_f$$

where: s_e = standard error of the estimate; s_f = standard error of the forecast; X_f = the value of X you are forecasting at. Better fit, bigger n , and X_f closer to $\bar{X}$ all shrink s_f .

FORECAST ROA AT CAPEX = 6% · SAME SIX COMPANIES

$$\hat{Y} = 4.875 + 1.25X; s_e = 3.4596; \bar{X} = 6.1; \sum (X - \bar{X})^2 = 122.64; n = 6; t^*_{0.025,4} = 2.776.$$

1. Point forecast: $\hat{Y}_f = 4.875 + (1.25 \times 6) = 12.375\%$.
2. Forecast SE: $s_f = 3.4596 \times \sqrt{1 + 1/6 + (6 - 6.1)^2 / 122.64} = 3.4596 \times \sqrt{1.16675} = 3.7369$.
3. Margin: $2.776 \times 3.7369 = 10.3737$.
4. 95% PI: $12.375 \pm 10.374 \rightarrow [2.00\%, 22.75\%]$.

Point forecast 12.4%; 95% interval [2.0%, 22.7%] — and this is the NARROWEST it gets, since $X_f \approx \bar{X}$.

When the relationship is not linear, take logs — log-lin gives a growth rate, lin-log a level response, log-log an elasticity

Simple linear regression is still doing the work — you are just feeding it transformed variables. The model stays linear in its PARAMETERS.

Form	Equation	What the slope means	R ² : plain vs transformed
Log-lin	$\ln Y = b_0 + b_1 X$	%Δ in Y per unit Δ in X (growth)	Revenues on time: 0.948 → 0.981
Lin-log	$Y = b_0 + b_1 \ln X$	Unit Δ in Y per %Δ in X	Margin on units: 0.551 → 0.972
Log-log	$\ln Y = b_0 + b_1 \ln X$	An ELASTICITY: %Δ Y per %Δ X	Revenues on ads: 0.209 → 0.849

COMPARING MODELS FAIRLY

You may NOT compare R² or F across models whose dependent variable is in a different form — lnY and Y are not the same quantity. When Y is unchanged (lin-log), s_e is comparable: 2.2528 falls to 0.5659. Otherwise, judge by the residual plot.

FORECASTING FROM A LOG MODEL

The model predicts ln Y, not Y. If $\ln Y = -7 + 2X$ and $X = 2.5$, the prediction is $\ln Y = -2$, so $\hat{Y} = e^{-2} = 0.1353$. Forgetting the antilog is the most common error in the whole learning module.

Source: CFA Institute, Curriculum Vol 1, Learning Module 10, §7 Functional Forms, Exhibits 38–43, pp. 304–309.

SECTION 06

Big data & ML wrap

Big data is defined by three Vs — volume, velocity, variety — plus a fourth, veracity, the moment the data is used for inference

Traditional data — prices, filings, economic releases — has not gone away. Alternative data has simply grown around it far faster, and with far weaker provenance.

VOLUME

Megabyte → gigabyte → terabyte → petabyte. Millions to billions of data points in a single dataset.

VELOCITY

Real-time and near-real-time ticks, clickstreams and sensor feeds. Batch and periodic updates still exist but are no longer the default.

VARIETY

Structured (SQL tables), semi-structured (HTML, JSON), unstructured (video, audio, free text). The pipeline has to handle all three.

THE FOURTH V • VERACITY

Scraped reviews, unverified posts, mislabelled satellite imagery, sensor drift. Big data amplifies the oldest problem in the discipline — separating quality from quantity — and adds legal and ethical exposure when the data was never meant to be public. Assessing the source is now a first-order analytical step, not a footnote.

¹ Alternative data has three sources: individuals, business processes and sensors (satellites, geolocation, the Internet of Things).

Source: CFA Institute, Curriculum Vol 1, Learning Module 11, §2 Fintech and Big Data, Exhibits 1–2, pp. 330–334.

Machine learning splits by whether you have labels — supervised, unsupervised, or deep learning through multi-layer neural nets

"Find the pattern, apply the pattern" is the curriculum's one-line summary of ML. It needs enormous training data, which is precisely why big data and ML arrived together.

Class	Training data	Example use in investing
Supervised learning	Inputs AND labelled outputs (a target)	Forecast next-day return direction; predict credit losses
Unsupervised learning	Inputs only – no labels, no target	Cluster stocks into peer groups; flag anomalous trades
Deep learning (neural nets)	Either – hidden layers build the features	NLP on earnings calls; image recognition on satellite data

YOUR REGRESSION WAS ALREADY SUPERVISED LEARNING

Simple linear regression is the simplest supervised-learning algorithm there is: labelled (X, Y) pairs, a model that learns a mapping, and an out-of-sample question. ML generalises it to non-linear forms and thousands of features — but the bias-variance trade-off, the train / validate / test discipline and the "is this real or is this noise" question are identical. The cost of the generality is interpretability: an ML model is often a black box, and a regression coefficient never is.

Source: CFA Institute, Curriculum Vol 1, Learning Module 11, §3 Advanced Analytical Tools: AI and Machine Learning, pp. 334–337.

Overfitting is the cardinal sin — a model that fits the training data perfectly and fails on data it has never seen

Split the data three ways: train to fit, validate to tune, test to measure honest out-of-sample performance. Touch the test set once, at the end, or it stops being a test. (The 60/20/20 split below is market convention, not curriculum.)

OVERFITTING · TOO COMPLEX

The model treats noise as if it were signal — "overtrained". Near-perfect in training, poor out of sample. Typical causes: too many parameters, too few observations, no regularisation. It discovers relationships that were never there.

UNDERFITTING · TOO SIMPLE

The model treats real signal as if it were noise. Poor in training AND poor out of sample. Typical cause: a functional form too restrictive for the data — a straight line through an obviously curved relationship.

Split	Role	Typical share
Training set	Fit the model parameters	60%
Validation set	Tune hyperparameters; choose between model families	20%
Test set	Report honest out-of-sample performance — once	20%

Source: CFA Institute, Curriculum Vol 1, Learning Module 11, §3 Advanced Analytical Tools, pp. 335–336.

Deck 3 — Sampling, Hypothesis Testing & Simple Linear Regression

- Probability sampling (simple, systematic, stratified, cluster) gives every element a known selection chance; non-probability sampling (convenience, judgmental) trades accuracy for speed.

- CI for $\mu = \bar{X} \pm t^* \cdot s/\sqrt{n}$, $df = n - 1$. It does NOT say there is a 95% chance μ is inside; the randomness lives in the interval, not in μ .

- Type I is α , Type II is β , power is $1 - \beta$. Only a bigger n improves both. Fix the number of tails from the QUESTION, before the data.

- OLS: $\hat{\beta}_1 = \text{cov}(X,Y)/\text{var}(X)$, $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1\bar{X}$. $R^2 = \text{SSR}/\text{SST} = r^2$ in SLR; slope $t = (\hat{\beta}_1 - B_1)/s_{\hat{\beta}_1}$ with $df = n - 2$, and $t^2 = F$. Test against 1, not just 0.

- CLT: $\bar{X} \approx N(\mu, \sigma^2/n)$ for ANY finite-variance population once n is large. Standard error falls with $1/\sqrt{n}$ — quadruple the data to halve the error.

- Six steps every time: state H_0/H_a , pick the statistic, set α , write the rule, compute, decide. $p \leq \alpha \Leftrightarrow \text{reject}$. Never "accept" the null.

- Match test to question: t (means), χ^2 (a variance, and independence), F (two variances), t on r (correlation, Pearson or Spearman rank).

- Prediction intervals widen away from $\bar{X}$. Take logs when the relation curves: log-log slopes are elasticities. Big data: 3 Vs + veracity; ML is supervised / unsupervised / deep; overfitting is the default failure.